

AYUSH KUMAR GUPTA

Computer Science & Data Analytics | IIT Patna

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PROFESSIONAL SUMMARY

Computer Science undergraduate at IIT Patna with strong experience in secure mobile app development, Post-Quantum Cryptography (PQC), AI-powered systems, XR/VR/AR applications, and real-time software solutions. Skilled in Flutter, Firebase, Unity, Python, REST APIs, Blockchain-integrated systems, and automation frameworks. Hackathon winner with proven leadership, research-oriented development, and cybersecurity expertise.

EDUCATION

Indian Institute of Technology Patna — B.Sc. Computer Science & Data Analytics

Graduation: 2027 | CPI: **9.12** | Latest SPI: **9.26**

CISCE Board — Class 12: 93% | Class 10: 91%

TECHNICAL SKILLS

Languages	Dart, Python, Java, JavaScript, HTML5/CSS3, C#
Mobile & XR Dev	Flutter, Unity, VR/AR Development (Image & Surface Tracking), Cross-platform Mobile
AI & Backend	Firebase (Auth, Firestore, Storage), REST APIs, Flask, FastAPI, TFLite, YOLOv5/YOLOv8
Security & Crypto	CRYSTALS-Kyber-512, AES-256-GCM, OQS Library, OTP Auth, Hybrid Encryption, Secure APIs
Blockchain	Solidity, Ethereum Sepolia, IPFS, web3dart
Simulation	SOFA Framework, Soft-Body Simulation, Mesh Deformation, Physics-Based Interaction
Automation	Selenium, Playwright, Browser Automation, Web Testing
Tools	Git & GitHub, Android Studio, Unity Editor, VS Code, Postman, Firebase Console

PROJECTS

Quantum-Safe Self-Destructing File Sharing System

Research Internship – NIT Agartala | 2025

- Built a full-stack Post-Quantum Secure communication platform using CRYSTALS-Kyber-512 and AES-256-GCM encryption.
- Integrated decentralized IPFS storage and Ethereum smart contracts for verifiable one-time secure file access.
- Developed a Flutter mobile application with secure private-key management and blockchain state validation.
- Built a Flask-based PQC server using the Open Quantum Safe (OQS) library for encapsulation and decapsulation workflows.
- Designed a self-destructing secure file-sharing architecture focused on privacy-preserving communication systems.

AI Smart Helmet – Real-Time Traffic Signal Detection

Hackathon Winner – IIT Patna | 2025

- Led mobile application development in a 5-member hackathon team for AI-assisted traffic signal detection.
- Integrated YOLOv5/YOLOv8-based real-time traffic signal recognition with rider alert systems.
- Developed a Flutter interface with live camera integration, voice alerts, and rider-focused UI components.
- Worked with FastAPI backend services and explored TensorFlow Lite for on-device inference optimization.
- Improved two-wheeler rider safety through real-time AI-powered road assistance systems.

SecureShare+ / Secure Print – Privacy-Focused File Sharing and Printing System

Independent Project | 2025

- Developed a one-time secure document sharing and printing platform for cyber cafes and public systems.

- Implemented auto-expiring secure links and zero-knowledge access workflows for privacy-focused access control.
- Applied quantum-safe encryption principles to prevent unauthorized access and file retention.
- Integrated Firebase Authentication, Firestore Security Rules, and secure file-access management systems.
- Designed client-side secure access workflows for temporary and privacy-preserving document sharing.
- Automated the printing process without letting the cafes system save or even see the document on the system.

Neighbour Nest – Community Commerce & Assistance Platform

Independent Project | 2024–2025

- Built a cross-platform Flutter application connecting users with nearby stores, clinics, and local service providers.
- Implemented real-time service listings, appointment booking, order placement, and notification systems.
- Added OTP-based authentication, multilingual support, and Firebase backend integration for scalable access.
- Developed community-focused features supporting local commerce, emergency assistance, and neighborhood connectivity.
- Optimized real-time Firestore data handling for responsive user interaction and dynamic updates.
- Published the application for real-world usage, gaining hands-on experience in deployment, backend management, and production workflows.

Air Quality Monitor App – Real-Time Pollution Tracker

Independent Project | 2025

- Developed a Flutter-based mobile application for monitoring real-time Air Quality Index (AQI) data using external APIs and ML models (LSTM).
- Built interactive dashboards with pollution indicators, AQI visualization, and forecast tracking systems.
- Implemented real-time unsafe-level alerts and health notifications for pollution monitoring.
- Added multi-location comparison features to help users track environmental conditions effectively.
- Focused on responsive UI design and efficient real-time API integration workflows.

VR-Based Liver Surgery Simulation using Unity + SOFA Framework

Ongoing Research Project | 2026

- Developing a VR-based soft-body liver surgery simulation system using Unity and the SOFA Framework.
- Implementing real-time mesh deformation, soft-tissue interaction, and physics-based collision handling systems.
- Integrating VR interaction mechanisms for surgical tool manipulation and immersive medical training environments.
- Exploring GPU-accelerated simulation workflows for realistic tissue behavior and low-latency rendering.
- Researching healthcare-focused XR technologies for surgical education and procedural simulation systems.

AR Interactive Applications using Unity

Ongoing Project | 2026

- Developed mobile augmented reality applications in Unity using image target and surface target tracking techniques.
- Implemented interactive 3D object placement, real-time AR visualization, and touch-based user interactions for smartphones.
- Optimized mobile XR performance and responsive AR workflows for smooth real-time user experiences.

Mobile Game Development using Unity

Ongoing Project | 2026

- Developed interactive 2D and 3D mobile game prototypes using Unity and C# scripting.
- Implemented gameplay mechanics, touch controls, collision systems, physics-based interactions, and UI components.
- Worked on real-time rendering, scene management, and mobile gameplay optimization for responsive user experiences.

ACHIEVEMENTS

- Hackathon Winner — IIT Patna (2025) for AI-powered traffic safety solution
- Research Internship — NIT Agartala (Post-Quantum Cryptography Project)
- Academic Excellence — 90%+ in both Class 10 and Class 12 (CISCE Board)
- Basketball Team Captain — led school team to multiple podium finishes in regionals tournaments
- Built multiple independent Flutter/Firebase applications and security-focused systems
- Appistry (App Development) Club Sub Coordinator – IIT Patna

RESEARCH & DEVELOPMENT INTERESTS

Quantum Security & Cryptography • Cybersecurity • XR/VR Simulation • AR Applications • AI & Computer Vision • Blockchain Security • Soft-Body Physics • Real-Time Systems